

Vpc creation

aws

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Peering connections

VPC > Your VPCs > vpc-0a13b9871f690a1fb

vpc-0a13b9871f690a1fb / nat-vpc

Actions

Details Info

VPC ID vpc-0a13b9871f690a1fb	State Available	DNS hostnames Disabled	DNS resolution Enabled
Tenancy Default	DHCP option set dopt-03d3fe889f95bed76	Main route table rtb-036d93f2c22a82910	Main network ACL acl-0cb52b90c9523bd62
Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -	IPv6 CIDR (Network border group) -
Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 600627341010	

Resource map

CIDRs

Flow logs

Tags

Integrations

Resource map Info

VPC Show details

Subnets (2)

Route tables (3)

Net

Creating IGW and attach to the VPC

[VPC](#) > [Internet gateways](#) > igw-09a7119c504ae0102

igw-09a7119c504ae0102 / nat-igw

Actions ▼

Details [Info](#)

Internet gateway ID

 igw-09a7119c504ae0102

State

 Attached

VPC ID

[vpc-0a13b9871f690a1fb](#) | [nat-vpc](#)

Owner

 600627341010

Tags

Manage tags

 Search tags

< 1 > 

Key

Value

Name

nat-igw

Created pub and pri subnets

Subnets (8) [Info](#)

Last updated
43 minutes ago



Actions ▼

Create subnet

Find resources by attribute or tag

< 1 > ⚙

☐	Name ▼	Subnet ID ▼	State ▼	VPC ▼	IPv4 CIDR ▼
☐	-	subnet-00c383009730e6bce	✓ Available	vpc-02ca2da100d130b4a	172.31.0.0/20
☐	-	subnet-01c75b4f9fbfaf24d	✓ Available	vpc-02ca2daf68d136b4a	172.31.0.0/20
☐	pub-sub	subnet-097dac7ec8c7f8a0b	✓ Available	vpc-0a13b9871f690a1fb nat-v...	10.0.1.0/24
☐	pri-sub	subnet-0972f644c6ca50e45	✓ Available	vpc-0a13b9871f690a1fb nat-v...	10.0.2.0/24

Select a subnet

Created pub and pri route table

Route tables (1/4) [Info](#)

Last updated 26 minutes ago

Actions ▾

Create route table

< 1 >

	Name ▾	Route table ID ▾	Explicit subnet associ... ▾	Edge associations ▾	Main ▾	VPC
☐	-	rtb-036d93f2c22a82910	-	-	Yes	vpc-0a13b9871f690a1f
☐	pub-rt	rtb-0bf0d3a024b8b288c	subnet-097dac7ec8c7f8a...	-	No	vpc-0a13b9871f690a1f
☒	pri-rt	rtb-0badc754f58aa94d8	subnet-0972f644c6ca50e...	-	No	vpc-0a13b9871f690a1f

rtb-0badc754f58aa94d8 / pri-rt

Details	Routes	Subnet associations	Edge associations	Route propagation	Tags
---------	--------	---------------------	-------------------	-------------------	------

Details			
Route table ID rtb-0badc754f58aa94d8	Main No	Explicit subnet associations subnet-0972f644c6ca50e45 / pri-sub	Edge associations -
VPC vpc-0a13b9871f690a1fb nat-vpc	Owner ID 600627341010		

Created NAT-GATEWAY

VPC > NAT gateways > nat-0965ff9e8e002af43

nat-0965ff9e8e002af43 / nat-gw

Actions ▼

Details

NAT gateway ID nat-0965ff9e8e002af43	Connectivity type Public	State Available	State message Info
NAT gateway ARN arn:aws:ec2:us-east-1:600627341010:natgateway/nat-0965ff9e8e002af43	Primary public IPv4 address 52.203.61.190	Primary private IPv4 address 10.0.1.54	Primary network interface ID eni-05767a6ae8095e2b5
VPC vpc-0a13b9871f690a1fb / nat-vpc	Subnet subnet-097dac7ec8c7f8a0b / pub-sub	Created Monday, October 28, 2024 at 11:50:58 GMT+5:30	Deleted -

Secondary IPv4 addresses | Monitoring | Tags

Secondary IPv4 addresses

Refresh Edit secondary IPv4 address associations

Search

< 1 > ⚙

Private IPv4 address ▼	Allocation ID ▼	Association ID ▼	Public IPv4 address
------------------------	-----------------	------------------	---------------------

Secondary IPv4 addresses are not available for this nat gateway.

Created two instance with pub and pri subnets

Instances (2) [Info](#)

Last updated less than a minute ago

Connect

Instance state ▼

Actions ▼

Launch instances ▼

All states ▼

Instance state = running X

Clear filters

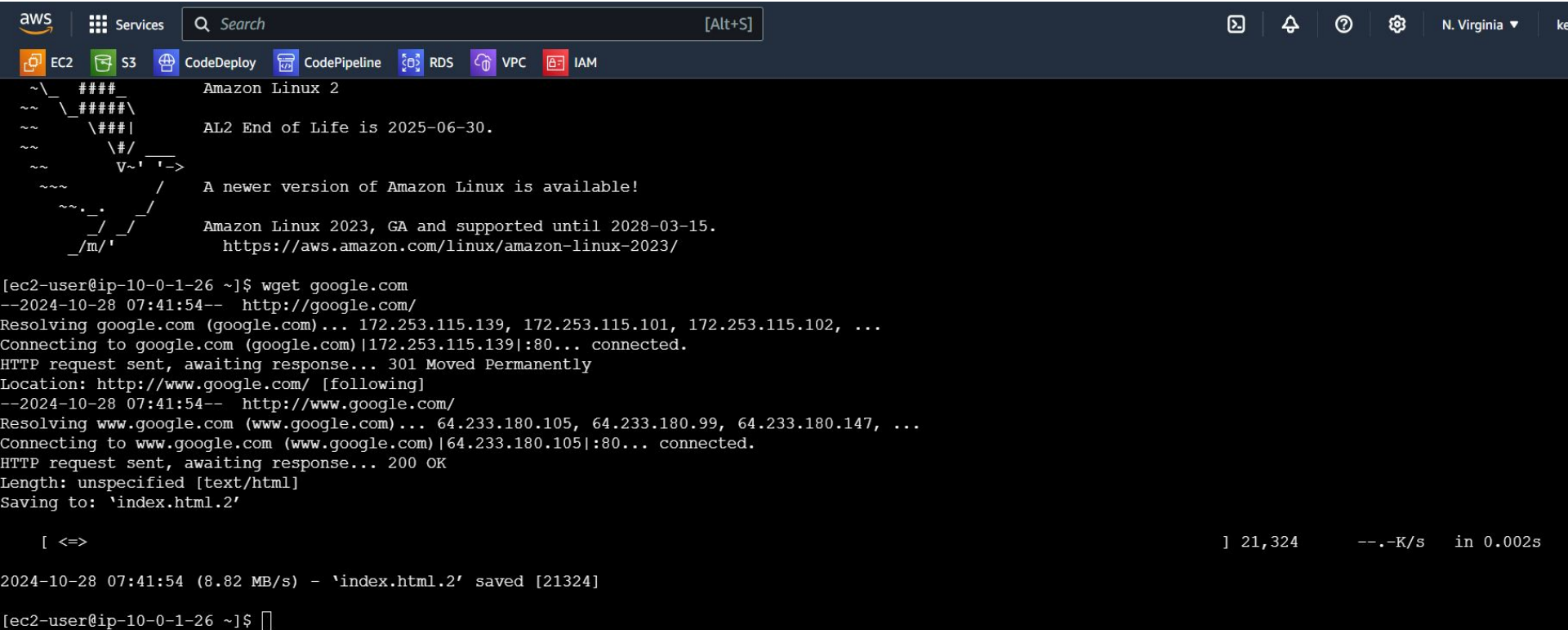
< 1 >

☐	Name	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm status	Availability Zone ▼	Public IP
☐	private-ser	i-0fbd9f3a638f2b706	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	-
☐	public-ser	i-01bc1a0e56f47cddf	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	-

Select an instance

=

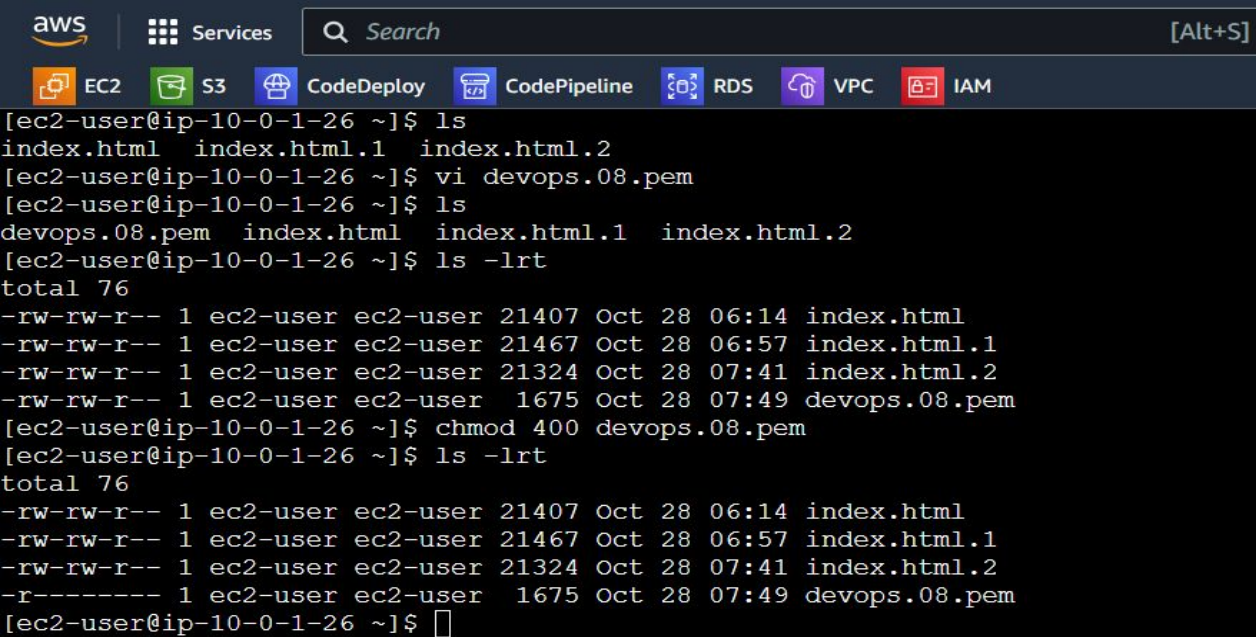
Pub server got internet bcoz of internet gateway



i-01bc1a0e56f47cddf (public-ser)

PublicIPs: 44.201.11.136 PrivateIPs: 10.0.1.26

After creating NATGATEWAY i couldn't establish an internet connection to private sever(pri-sub),so i create and save my pem file and also changed the permission.



The image shows a screenshot of the AWS Management Console interface at the top, with a dark blue header bar containing the AWS logo, a 'Services' menu, a search bar, and a keyboard shortcut '[Alt+S]'. Below the header, a row of service icons is visible: EC2, S3, CodeDeploy, CodePipeline, RDS, VPC, and IAM. The main area of the screenshot is a terminal window with a black background and white text, showing a series of commands and their outputs. The terminal session starts with a user prompt '[ec2-user@ip-10-0-1-26 ~]\$' followed by 'ls', which lists 'index.html', 'index.html.1', and 'index.html.2'. The user then runs 'vi devops.08.pem', followed by another 'ls' command showing 'devops.08.pem', 'index.html', 'index.html.1', and 'index.html.2'. The user then runs 'ls -lrt', which shows a directory listing with permissions, user, group, size, date, and filename. The permissions for 'devops.08.pem' are shown as '-rw-rw-r--'. The user then runs 'chmod 400 devops.08.pem', and finally runs 'ls -lrt' again, where the permissions for 'devops.08.pem' are now shown as '-r-----'. The terminal session ends with a prompt '[ec2-user@ip-10-0-1-26 ~]\$' and a cursor.

```
[ec2-user@ip-10-0-1-26 ~]$ ls
index.html  index.html.1  index.html.2
[ec2-user@ip-10-0-1-26 ~]$ vi devops.08.pem
[ec2-user@ip-10-0-1-26 ~]$ ls
devops.08.pem  index.html  index.html.1  index.html.2
[ec2-user@ip-10-0-1-26 ~]$ ls -lrt
total 76
-rw-rw-r-- 1 ec2-user ec2-user 21407 Oct 28 06:14 index.html
-rw-rw-r-- 1 ec2-user ec2-user 21467 Oct 28 06:57 index.html.1
-rw-rw-r-- 1 ec2-user ec2-user 21324 Oct 28 07:41 index.html.2
-rw-rw-r-- 1 ec2-user ec2-user  1675 Oct 28 07:49 devops.08.pem
[ec2-user@ip-10-0-1-26 ~]$ chmod 400 devops.08.pem
[ec2-user@ip-10-0-1-26 ~]$ ls -lrt
total 76
-rw-rw-r-- 1 ec2-user ec2-user 21407 Oct 28 06:14 index.html
-rw-rw-r-- 1 ec2-user ec2-user 21467 Oct 28 06:57 index.html.1
-rw-rw-r-- 1 ec2-user ec2-user 21324 Oct 28 07:41 index.html.2
-r----- 1 ec2-user ec2-user  1675 Oct 28 07:49 devops.08.pem
[ec2-user@ip-10-0-1-26 ~]$
```


With help of NAT GATEWAY i established an internet connection to my private server.(jumping pub-sub to pri-sub)

```

aws Services Search [Alt+S]
EC2 S3 CodeDeploy CodePipeline RDS VPC IAM
[ec2-user@ip-10-0-1-26 ~]$ ssh -i "devops.08.pem" ec2-user@10.0.2.10
#
~\  ##### Amazon Linux 2
~~ \ #####\
~~  \###| AL2 End of Life is 2025-06-30.
~~   \#/
~~    V~' '->
~~~
~~~_./ / A newer version of Amazon Linux is available!
~~_./ /
_/m/' / Amazon Linux 2023, GA and supported until 2028-03-15.
      https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-10-0-2-10 ~]$ wget google.com
--2024-10-28 08:09:20-- http://google.com/
Resolving google.com (google.com)... 217.160.0.201
Connecting to google.com (google.com)|217.160.0.201|:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: http://www.google.com/ [following]
--2024-10-28 08:09:21-- http://www.google.com/
Resolving www.google.com (www.google.com)... 217.160.0.201
Reusing existing connection to google.com:80.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'

[ <=> ] 23,343 --.-K/s in 0.1s

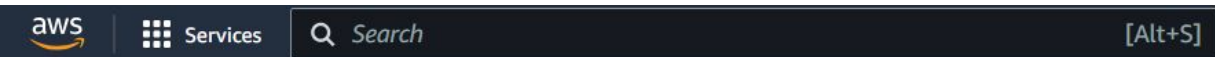
2024-10-28 08:09:21 (179 KB/s) - 'index.html' saved [23343]

```

i-01bc1a0e56f47cddf (public-ser)

PublicIPs: 44.201.11.136 PrivateIPs: 10.0.1.26

Again back from private server to public server



```
~~ \###| AL2 End of Life is 2025-06-30.
```

```
~~ \#/
~~ V~' '->
```

```
~~~ / A newer version of Amazon Linux is available!
```

```
~~~.-. /
    / Amazon Linux 2023, GA and supported until 2028-03-15.
  /m/' - https://aws.amazon.com/linux/amazon-linux-2023/
```

```
[ec2-user@ip-10-0-2-10 ~]$ wget goole.com
--2024-10-28 08:09:20-- http://goole.com/
Resolving goole.com (goole.com)... 217.160.0.201
Connecting to goole.com (goole.com)|217.160.0.201|:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: http://www.goole.com/ [following]
--2024-10-28 08:09:21-- http://www.goole.com/
Resolving www.goole.com (www.goole.com)... 217.160.0.201
Reusing existing connection to goole.com:80.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'index.html'
```

```
[ <=>
```

```
2024-10-28 08:09:21 (179 KB/s) - 'index.html' saved [23343]
```

```
[ec2-user@ip-10-0-2-10 ~]$ logout
Connection to 10.0.2.10 closed.
```